Installation and Setup Guide

Project: Microservice Monitor Ecosystem

Prepared by: Akın Mert Ak,Muhammed Şükrü Çiçek , Hasan Çeşmeci

1. Introduction

This document provides step-by-step instructions on how to install, configure, and run the **Microservice Monitor** project locally or on a server. The system integrates a microservice architecture consisting of User, Product, and Order services, Jenkins CI/CD pipelines, and Nagios monitoring infrastructure within Docker containers.

2. System Requirements

Ensure the following components are installed on your system for the application to run smoothly:

- **Docker Desktop:** For containerization operations (v20.10+ recommended).
- **Docker Compose:** For multi-container orchestration (v2.0+ recommended).
- **Git:** To clone the project repository.
- **Internet Connection:** Required to download Docker images and dependencies.

3. Installation Steps

3.1. Downloading the Project

First, open your terminal and clone the project from GitHub:

```
git clone https://github.com/mscicek/microservice-monitor.git
cd microservice-monitor
```

3.2. Building Docker Containers

Use the following command to build all services and install dependencies:

```
docker-compose build --no-cache
```

3.3. Running the System

To start the services in the background (detached mode), run this command in the root directory:

```
docker-compose up -d
```

After executing this command, you should see 6 containers (postgres, user-service, product-service, order-service, jenkins, nagios) in the "Running" state on Docker Desktop.

4. Service Access Information

Service Name	URL Address	Description / Credentials
User Service API	http://localhost:8001/docs	User management via Swagger UI
Product Service API	http://localhost:8002/docs	Product and stock management via Swagger UI
Order Service API	http://localhost:8003/docs	Order processing and inter-service communication
Nagios Dashboard	http://localhost:8080/nagios	User: `nagiosadmin` / Password: `nagios`
Jenkins Automation	http://localhost:8090	Initial admin password required for first setup
PostgreSQL DB	localhost:5432	User: `admin` / Password: `password`

5. Monitoring Configuration

You can verify the configuration by running this command:

```
docker exec nagios /opt/nagios/bin/nagios -v /opt/nagios/etc/nagios.cfg
```

6. Jenkins CI/CD Setup

1. Navigate to `http://localhost:8090`.
2. Retrieve the initial admin password from the terminal using: `docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword`
3. Create a new item of type "Pipeline".
4. Copy the contents of the `Jenkinsfile` from the project and paste it into the Pipeline script section.
5. Click "Build Now" to trigger the first deployment and health check.

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